PROJECT REPORT

# SMART SILENT MODE RECOMMENDER

An Intelligent Calendar-Based Silent Mode Recommendation System


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# ABSTRACT

This project is a calendar-based application that gives suggestions about when a phone should be kept in Silent Mode. The user can add an event by entering its name, date, start time and end time. The application saves the event and shows recommendations for today, tomorrow and the next seven days. Events such as classes, lectures, meetings, seminars and examinations can be identified as important events. The application displays the suggested mode along with a reason. It is developed using Python and Streamlit, with a database service to store calendar events.

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# 1. INTRODUCTION

Mobile phones are useful for communication and daily activities. But calls and notifications can disturb people during classes, meetings, seminars and examinations. Sometimes users forget to put their phones in Silent Mode before these events.

This project helps users plan their phone sound settings by using a calendar. Users can save their events and view suggestions about Silent Mode for the scheduled dates.

## 1.1 Problem

Changing the phone sound setting manually depends on the user remembering every event. This application makes it easier by showing important events and giving a Silent Mode suggestion.

## 1.2 Proposed System

The application provides options to add events, view saved events and delete events. It also displays recommendations for the current day, the next day and the upcoming seven days.

# 2. OBJECTIVES

- To create a simple calendar-based application.
- To allow users to add events with date and time.
- To save and display event details.
- To identify important events using keywords.
- To show Silent Mode or Normal Mode suggestions.
- To display recommendations for the upcoming seven days.
- To allow users to delete saved events.

# 3. SOFTWARE AND TOOLS USED

| Tool | Use |
|---|---|
| Python | Used to write the application and recommendation logic. |
| Streamlit | Used to create the web interface. |
| SQLite / database service | Used to store calendar events. |
| Pandas | Used for working with event data in table form. |
| datetime | Used to handle dates and times. |
| app.py | Contains the main application interface. |
| calendar_service.py | Contains database and event management functions. |
| recommender.py | Contains the logic used to suggest the phone mode. |

# 4. SYSTEM DESIGN

The project is divided into different parts. The Streamlit page is used by the user. The main Python file connects the page with the calendar service and recommendation function. The calendar service saves and retrieves events. The recommendation function checks the event details and returns a suggested mode.

| Part | Work done |
|---|---|
| User interface | Takes event details and displays results. |
| app.py | Connects the different parts of the application. |
| Calendar service | Adds, displays and deletes events. |
| Recommendation function | Checks event name, date and time. |

| Part | Work done |
| --- | --- |
| Database | Stores the event information. |

### 4.1 Main Features

**Add Event:** The user enters the event name, date, start time and end time and saves the event.

**Saved Events:** The application displays the saved events in a table with their details.

**Recommendations:** The application shows suggestions under different date headings.

**Manage Events:** The user can select and delete an event.

## 5. HOW THE APPLICATION WORKS

1. Open the Smart Silent Mode Recommender application.
2. Enter the event name.
3. Select the event date.
4. Select the start time and end time.
5. Click the Save Event button.
6. The event is stored in the database.
7. The application checks the saved events and displays the recommendations.

### 5.1 Recommendation Logic

The recommendation function checks the event title for important words. The keyword list used in the project includes class, meeting, seminar, exam, examination, presentation, lecture, interview and workshop. If an event title contains one of these words, the application treats it as an important event.

The event date and time are also checked. For an event scheduled today, the application can compare the current time with the event time. Events on future dates are shown in the upcoming recommendations. If no important event is found, Normal Mode can be suggested.

### 5.2 Basic Steps

1. Read the event details.
2. Check the event name for important keywords.
3. Check the event date.
4. Compare the time when required.
5. Return the suggested mode and reason.

# 6. DATABASE DETAILS

The application stores calendar event details in a database. The calendar service provides functions to create the database, add an event, get the saved events and delete an event.

| Field | Description |
|---|---|
| ID | Unique number for an event. |
| Event Name | Name entered by the user. |
| Date | Date of the event. |
| Start Time | Starting time of the event. |
| End Time | Ending time of the event. |

These fields match the event information shown in the Saved Events table in the supplied screenshots.

# 7. APPLICATION SCREENSHOTS

The following screenshots were captured from the working project application.

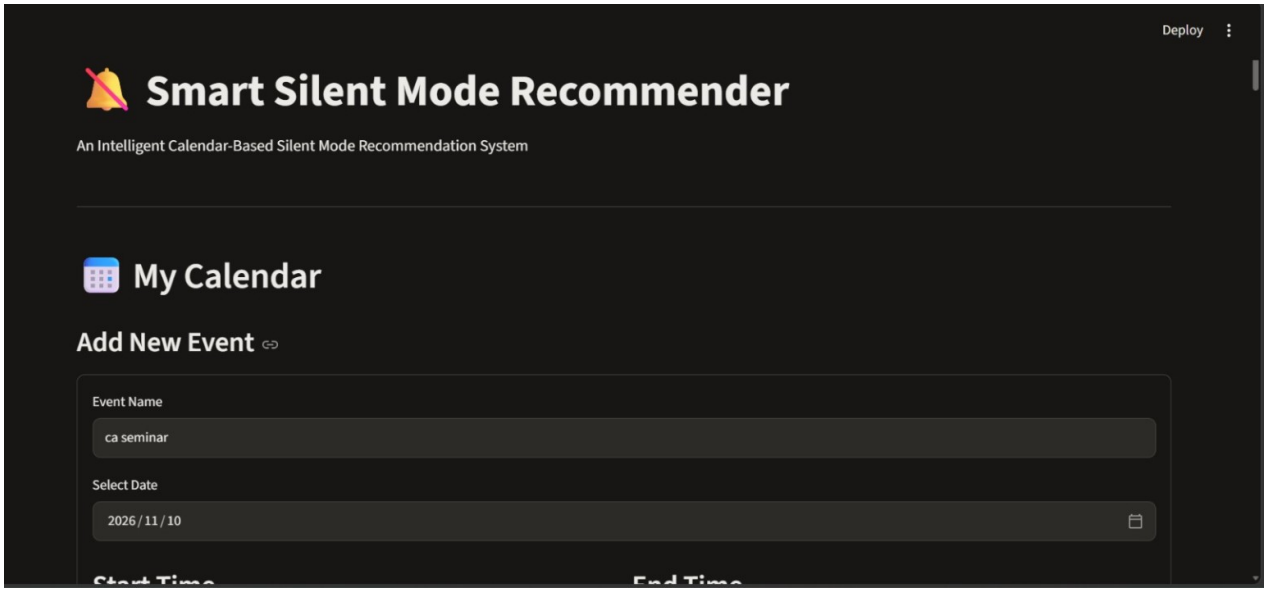
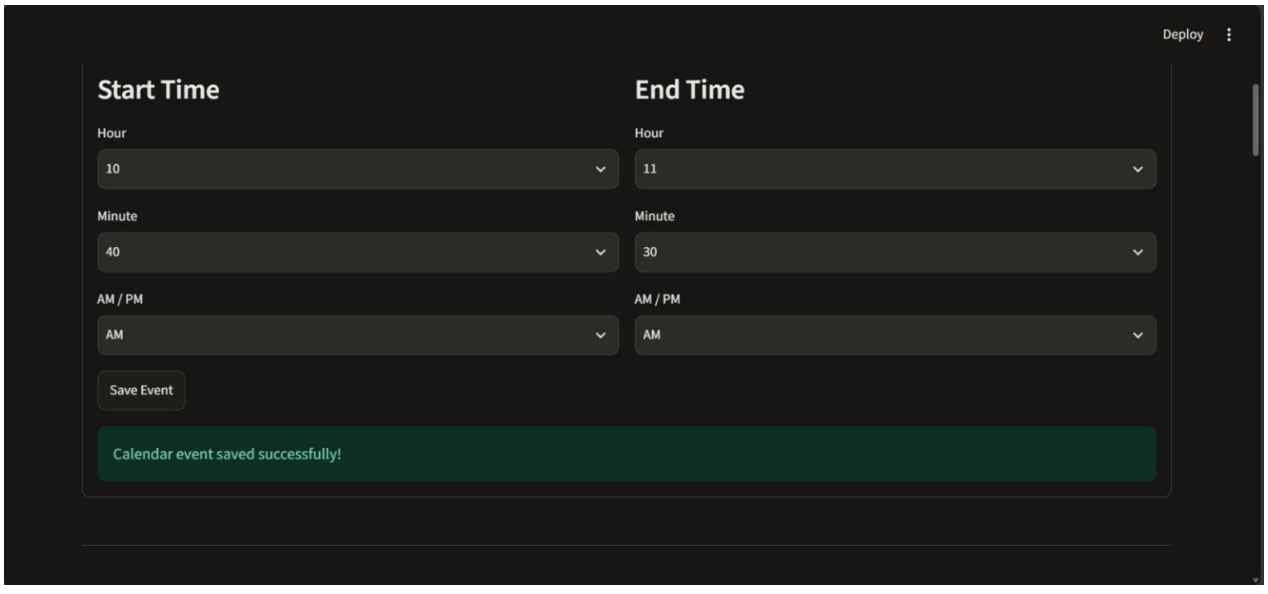



*Figure 1: Main page and Add New Event form*



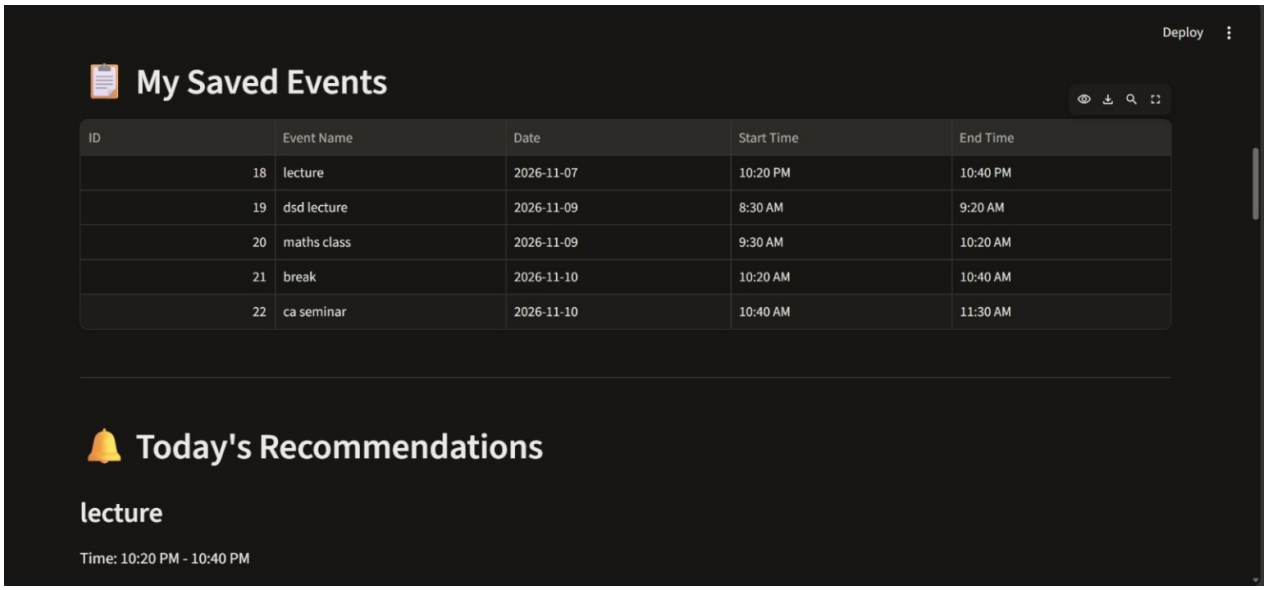



*Figure 3: Saved events and today's recommendation*

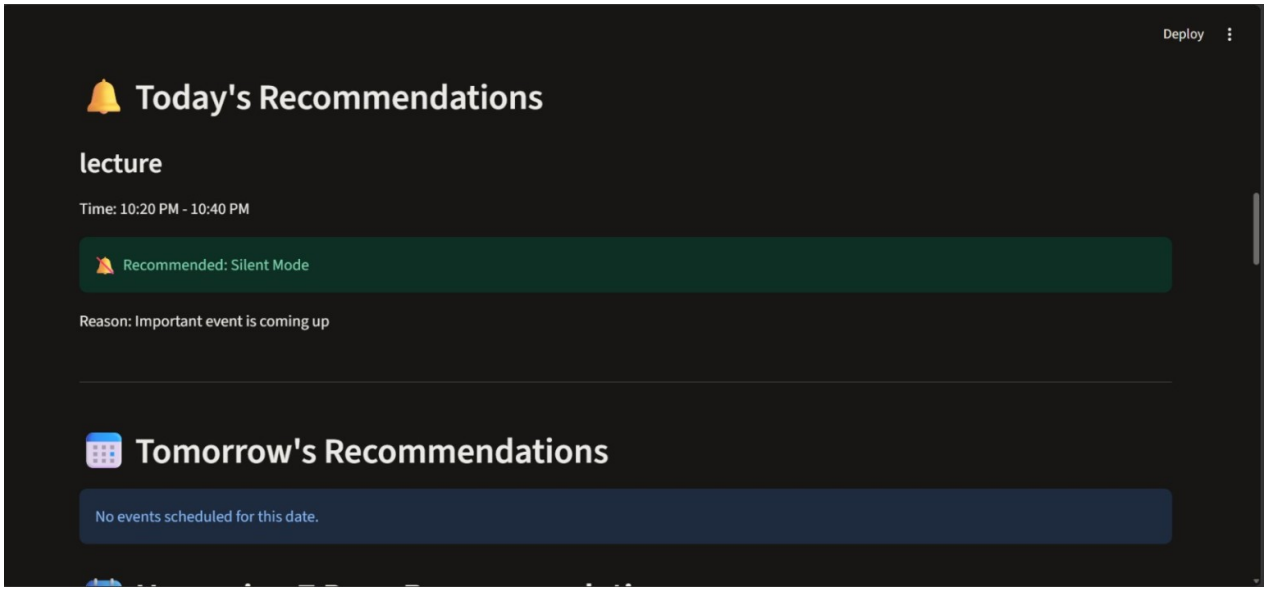



*Figure 4: Today's and tomorrow's recommendations*

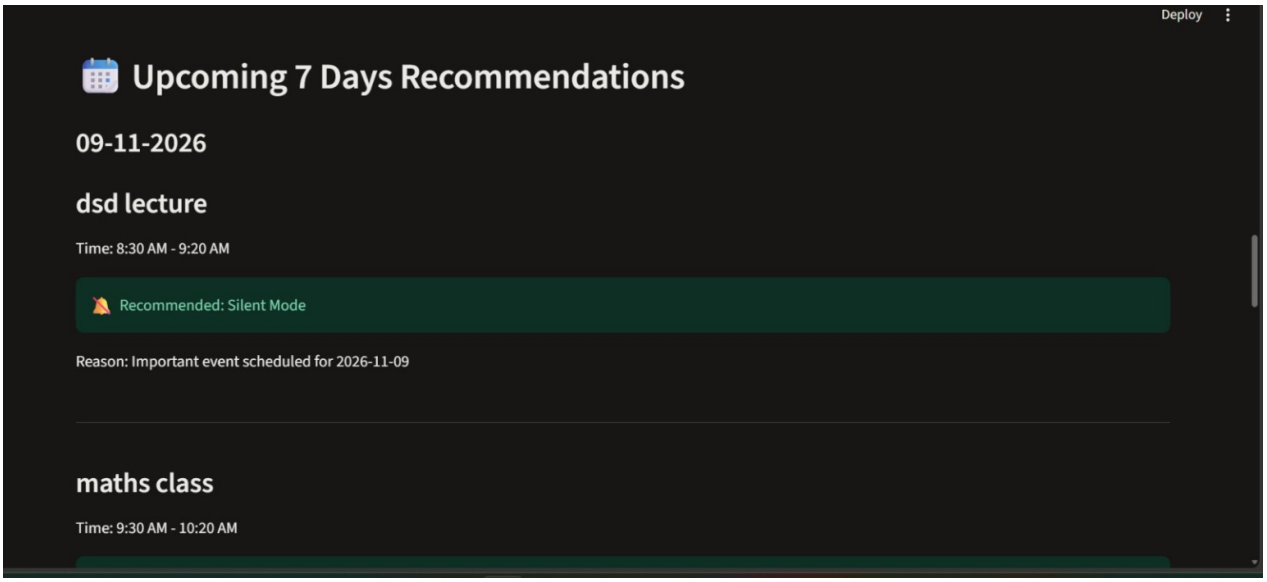



*Figure 5: Recommendations for the next seven days*

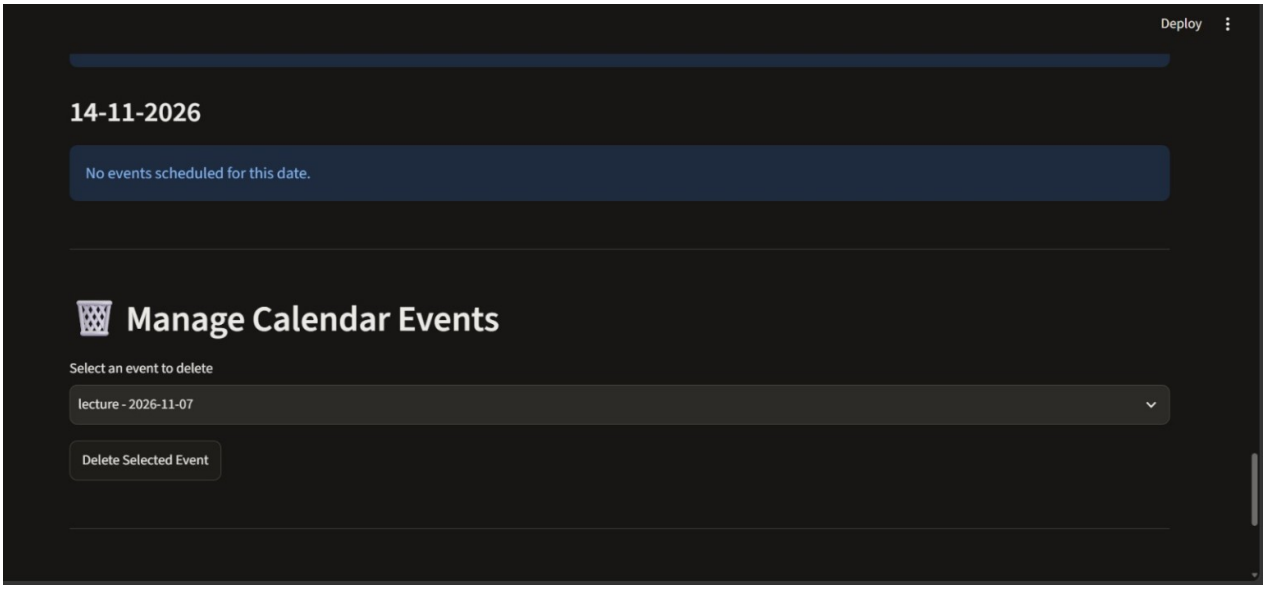



*Figure 6: Event management and a date without scheduled events*

# 8. CONCLUSION

The Smart Silent Mode Recommender project provides a simple way to manage events and view phone mode suggestions based on a calendar. Users can add events, see the saved event list, check recommendations for different dates and delete events when needed. The project uses Python, Streamlit and a database service. It helped demonstrate how event details and simple conditions can be used to give useful recommendations.

## RELATION TO COMPUTER ARCHITECTURE

The CPU processes event details and recommendation conditions. Memory holds data during execution, while the database stores events for later use. This connects the project to computer architecture.

# REFERENCES


1. Python Documentation · https://docs.python.org/3/
2. Streamlit Documentation · https://docs.streamlit.io/
3. Pandas Documentation · https://pandas.pydata.org/docs/
4. SQLite Documentation · https://www.sqlite.org/docs.html


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END OF REPORT